Conversely, VibeCheck best predicts preferences for writing-oriented prompts, as style is often more important for these open ended tasks.

To understand how user preferences for these vibes vary across task domains and contexts, we analyze separability scores and preference prediction coefficients for predefined vibes in Figure 3. For writing tasks, formality, humor, and expressive emotional content positively correlate with user preference, while these traits negatively correlate with STEM tasks, where logical rigor is the most influential on preference. Conversely, logical rigor has minimal impact on preferences for writing tasks. While our dataset does not directly compare individual judgments, treating STEM and writing task users as distinct groups provides preliminary evidence of task-specific preferences. Additionally, lower separability scores for STEM tasks indicate less stylistic divergence in model outputs for objective questions like coding and math, making model identity harder to predict, consistent with Table 2.

**Notable vibes for Llama-3 70B.** The top 10 vibes uncovered by VibeCheck (Figure 2) highlight Llama's use of formatting, willingness to engage with sensitive topics, less emphasis on ethics, and a conversational, humorous style. Finer-grained vibes include Llama's use of bold/italics to emphasize points and increased use of personal pronouns, with 'I,' 'we,' and 'you' appearing 3x more in Llama outputs than GPT/Claude conversations. The preference prediction coefficeients in Figure 2 show Chatbot Arena users tend to prefer outputs which are less focused on ethics, employ markdown and typographic emphasis to highlight key points, and employ humor to engage the user, all of which are vibes which llama possesses. We believe that this correlation between vibes and user preference can explain some of the discrepancy seen in llamas high ranking on the leaderboard in comparison to models like GPT-4 which often outperform Llama.

| **Vibe** (low -> high) | **Sep Score** [-0.4,0.4] | **PP Coef** [-0.5,0.5] | **Cohen** |
|---|---|---|---|
| **Language and Tone.** Professional, straightforward tone. -> Enthusiastic, friendly tone. | | | 0.51 |
| **Typographic Emphasis.** Minimal use of typographic emphasis, letting the text stand alone. -> Uses typographic emphasis like bold or italics to highlight key points. | | | 0.64 |
| **Interactivity.** Provides information passively without engaging the user. -> Encourages user interaction, such as posing questions or suggesting actions. | | | 0.44 |
| **Formatting Completeness.** Responses are minimally formatted, relying on plain text. -> Responses include comprehensive formatting, such as Markdown or additional stylistic elements. | | | 0.57 |
| **Examples and Illustrations.** Minimal examples. -> Provides multiple examples. | | | 0.61 |
| **Use of Humor.** Maintains a serious tone without humorous elements. -> Employs humor frequently to engage the reader. | | | 0.62 |
| **Use of Personal Pronouns.** Rarely or never uses personal pronouns. -> Frequently uses personal pronouns (I, we, you). | | | 0.32 |
| **Ethical Consideration.** Provides factual information without commenting on ethics. -> Offers ethical considerations in its responses. | | | 0.53 |
| **Humility.** Projects confidence and completeness without discussing limitations. -> Frequently acknowledges limitations in the response or areas of uncertainty. | | | 0.41 |
| **Formality Level.** Uses informal or conversational language. -> Uses formal language and expressions. | | | 0.45 |

Figure 2: **Comparing Llama-3-70b VS GPT-4 & Claude-3-Opus on Chatbot Arena.** Negative separability scores indicate Llama-3-70B aligns with the low (red) description, while negative preference coefficients show alignment with low descriptions is preferred. We see that Llama is more humorous, utilizes more formatting, provides more examples, and comments much less on ethics than GPT and Claude: all attributes which correlate positively with human preference.

## 6 Applications

We next apply VibeCheck to discover qualitative differences between models' behavior on three open-ended tasks: text summarization, math problem-solving, and image captioning. We use CNN/DailyMail (Hermann et al., 2015) for text summarization, MATH (Hendrycks et al., 2021b) with chain-of-thought prompting for problem-solving, and COCO for image captioning. For CNN and MATH we use cached model predictions downloaded from HELM (Liang et al., 2023) and intentionally choose models which are ranked similarly to each other, but when running LLM as a judge to get preference labels, one model is more heavily preferred. For captioning, we generate captions on a random subset of 1000 COCO images. The vibes for each application in Section G.

| Method | Overall | | | STEM | | | Writing | | |
|---|---|---|---|---|---|---|---|---|---|
| | M.M. | P.P. | C.K. | M.M. | P.P. | C.K. | M.M. | P.P. | C.K. |
| VibeCheck [1 iter] | 68.68 | 60.00 | 0.42 | 65.20 | 55.99 | 0.44 | 74.09 | 60.58 | 0.51 |
| VibeCheck [3 iter] | **80.34** | 59.34 | 0.46 | **68.71** | 57.31 | 0.45 | **77.19** | **62.04** | 0.49 |
| Predefined Vibes | 72.10 | **61.11** | 0.51 | 65.94 | **58.38** | 0.45 | 75.00 | 59.49 | 0.52 |

Table 2: **Comparing Llama-3 to GPT and Claude on Chatbot Arena.** We report Model Matching Accuracy (M.M.), Preference Prediction Accuracy (P.P.), and average Cohen's Kappa (C.K) for the full dataset (Overall) and STEM and Writing categories. VibeCheck achieves higher model matching accuracy than Predefined Vibes and similar preference prediction accuracy. VibeCheck obtains the largest improvements over predefined vibes in the writing category, suggesting that for open-ended prompts, model styles differ significantly, and style has a greater influence on preference.

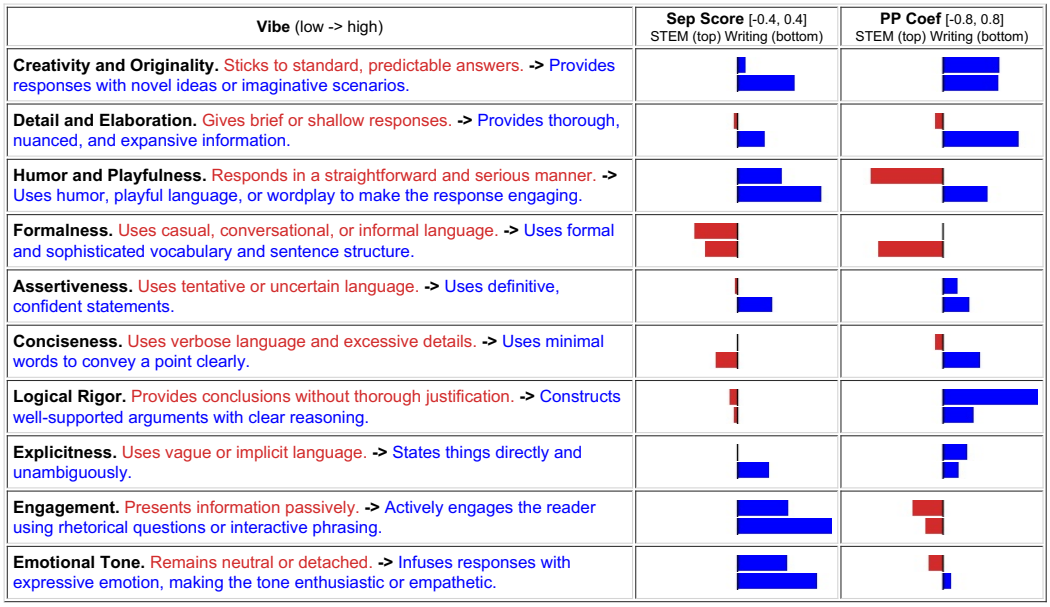


Figure 3: **Comparing user preference and separability across STEM and writing tasks.** On predefined list of vibes referenced in Table 2. Negative preference coefficients indicate a preference for low-description vibes, while negative separability scores show Llama responses align more with the low description than Claude or GPT responses. For writing tasks, detailed explanations, humor, and expressive emotion positively correlate with human preference, while these traits negatively correlate with STEM tasks. Conversely, logical rigor has a stronger positive impact on preference for STEM tasks. These trends are reflected in separability scores, with less separability on STEM tasks for vibes like humor and emotional tone, and more separability for logical rigor.

### 6.1 What do Different Models focus on When Summarizing?

We compare the summary styles of TNLG v2 (Smith et al., 2022) (530B) to Cohere's Command X large Beta (Inc., 2023) on the CNN/DailyMail dataset. While these models achieve a similar mean win rate on the HELM leaderboard, we see when using LLM as a preference judge, Command X has a win-rate of 71.12%. Looking at the top 5 vibes located in Figure 14, we find that (1) Command X clearly states an introduction and conclusion while TNLG utilizes choppy sentences without an either (2) Command provides specific examples or anecdotes to illustrate points and (3) Command is able to capture multiple viewpoints and emotional aspects of a story while TNLG is more objective. We see these qualities are positively correlated with human preference, which may explain the disparity between correctness metrics and preference metrics. With these vibes, we achieve a model matching accuracy of **71.29%** and a preference prediction accuracy of **61.42%**.

**Prompt:** Summarize the article in 3 sentences.

**Command X:** A giant filter-feeding 'lobster' took the place of whales 480 million years ago, a new fossil find has shown. The 6.5ft (two-metre) prehistoric creature was an early ancestors of modern crustaceans, insects and spiders. At this length it would have been comparable in size to athlete Usain Bolt. The remains were unearthed in Morocco and it is thought to be the oldest giant filter-feeder ever discovered.

**TNLGv2:** A giant filter-feeding 'lobster' took the place of whales 480 million years ago . 6.5ft (two-metre) prehistoric creature belonged to the family of anomalocaridids . It was a 'gentle giant' .

### 6.2 How do different LLMs solve math problems?

Objective tasks like math have a single final answer, but the way a model explains its thought process varies across models. We run VibeCheck on the MATH dataset (Hendrycks et al., 2021c) using chain-of-thought prompting to discover how GPT-4o and Llama-405b differ in their thought process and presentation. To reduce the variance seen from incorrect examples, we run VibeCheck only on the questions where both models answered correctly and aim to discover why GPT-4o is favored in 76% of conversations. Inspecting the top 5 vibes in Figure 4, we observe that Llama-405b organizes its responses under markdown headings, adopts a more conversational tone, and includes overly detailed step-by-step explanations, as illustrated below. Examining the coefficients of the preference prediction model, we find that a formal tone and frequent use of notation positively correlate with preference, while over-explaining the reasoning process negatively correlates with preference. These vibes achieve a model-matching accuracy of **97.09%** and a preference prediction accuracy of **72.79%**.

**Prompt:** Find the positive difference between 2(6) + 4(3) and 8(3+3)

**GPT-4o:**
To find the positive difference between 2(6) + 4(3) and 8(3+3), we first need to evaluate each expression separately.

First, evaluate 2(6) + 4(3)

$$2(6) + 4(3) = 2 \cdot 6 + 4 \cdot 3 = 12 + 12 = 24$$

Next, evaluate 8(3+3):

$$8(3 + 3) = 8 \cdot (3 + 3) = 8 \cdot 6 = 48$$

...

**Llama-405b:**
## Step 1: Evaluate the expression 2(6) + 4(3)
First, we need to evaluate the expression 2(6) + 4(3) by following the order of operations (PEMDAS/BODMAS).
We start by multiplying 2 and 6, which gives us 12. Then, we multiply 4 and 3, which gives us 12.
Finally, we add the two results together, giving us 12 + 12 = 24.

## Step 2: Evaluate the expression 8(3+3)
Next, we need to evaluate the expression 8(3+3) by following the order of operations (PEMDAS/BODMAS). We start by evaluating the expression inside the parentheses, which gives us 3 + 3 = 6. Then, we multiply 8 by 6, which gives us 48.
...

| **Vibe** (low -> high) | **Sep Score** [-0.9,0.9] | **PP Coef** [-0.6,0.6] | **Cohen** |
|---|---|---|---|
| **Explanation and Step-by-Step Detail.** Continuous narrative, no explicit step labels, less granularity. **->** Detailed steps, labels (e.g., '## Step 1'), granular breakdown. | | | 0.90 |
| **Structural Formatting.** Continuous prose without explicit structuring. **->** Uses headings, subheadings, numbered lists. | | | 0.70 |
| **Conciseness.** Extended discussions, unnecessary commentary, contains repetition. **->** Brief, to-the-point explanations, no unnecessary repetition. | | | 0.51 |
| **Efficiency of Steps.** Detailed intermediary steps, broader explanations. **->** Concise, straightforward solution steps. | | | 0.42 |
| **Mathematical Notation Use.** More written explanations, fewer symbols. **->** Frequent use of symbols, LaTeX/MathML formatting. | | | 0.33 |

Figure 4: **Top 5 vibes comparing GPT-4o to Llama-3-405B on MATH CoT.** Negative separability scores indicate GPT-4o aligns with the low (red) description, while negative preference coefficients show alignment with low descriptions is preferred. GPT-4o outputs contain more LaTex/MathML formatting which positively correlated with human preference while Llama-3-405B has very structured and overly-detailed responses, which is negatively correlated with preference.

### 6.3 What are VLM's Captioning Style?

Image captioning is one of the most popular use cases for Vision and Language models, but different captioning models focus on different image properties. We run VibeCheck on captions generated by GPT-4V (Chen et al., 2023; OpenAI, 2023) and Gemini-1.5-Flash (Reid et al., 2024) on 1000 COCO images and we find that GPT-4V uses more poetic language and structures its captions as a dynamic story, inferring the personality and emotions of the subjects in the image while Gemini sticks to more literal descriptions (Figure 16). The top 10 vibes generated by VibeCheck are able to achieve near